

MX17 — Thermal-window (> 1 ms) Trigger Optimization

SiPM wall $\times$ plastics: tagging X17 & IPC pairs, rejecting capture- γ background

Geant4 full-simulation study

2026-07-19 (*final surveyed geometry*)

The question

Goal: with the final geometry, decide how to trigger in the *thermal neutron* time window (arrival > 1 ms after the γ flash) to efficiently tag possible **X17** and **IPC** e^+e^- pairs while rejecting neutron-capture backgrounds.

Two trigger detectors per arm

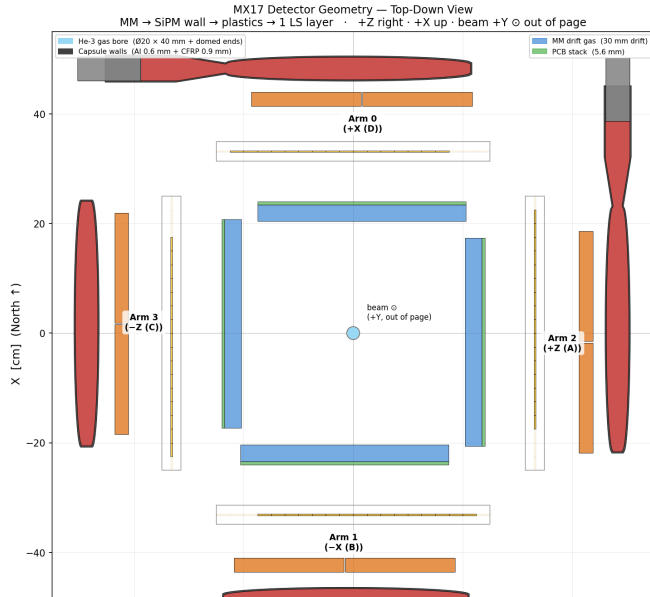
- ▶ **SiPM wall** — 16 thin (3 mm) scint bars; gives the 2-arm pair *topology*.
- ▶ **Plastics** — 2 thick (2.5 cm) bars; give *energy* (per-leg pulse height).

Design levers

- ▶ SiPM per-bar threshold a
- ▶ Plastic per-bar threshold b
- ▶ Coincidence topology (1 leg / 2 legs / ...)

Both thresholds are quoted in **MIP** units, so they transfer directly to hardware via the measured MIP peaks. 1 MIP = 458 keV (SiPM bar), 4.33 MeV (plastic).

Setup — 4-arm detector, top-down



Beam (+Y) into the page; four arms in the transverse plane.

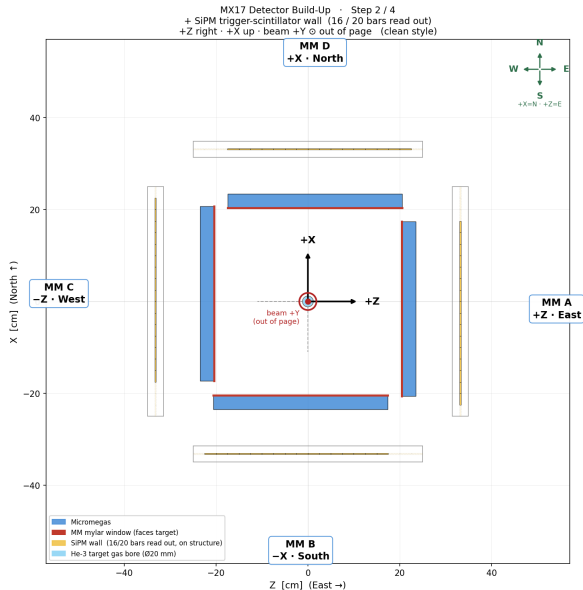
Stack, inside → out per arm:

1. Micromegas (tracking)
2. **SiPM wall** (trigger)
3. **plastics** (trigger)
4. liquid-scint layer

He-3 target at the centre:

Ø20 × 40 mm gas bore in an Al+CFRP capsule.

Setup — the trigger stack



SiPM wall (yellow): 50 cm span, 16 of 20 bars read out, bars run along the beam, array along the tangent. Centered on the mechanical structure.

Plastics (orange): two $20 \times 30 \times 2.5$ cm bars, tape+Al wrapped, behind the wall, centered on the Micromegas.

A leg “fires” when its SiPM bar *and* a plastic bar are both above threshold.

Campaigns (Geant4 11.2, FTFP_BERT_HP; final geometry; all files validated):

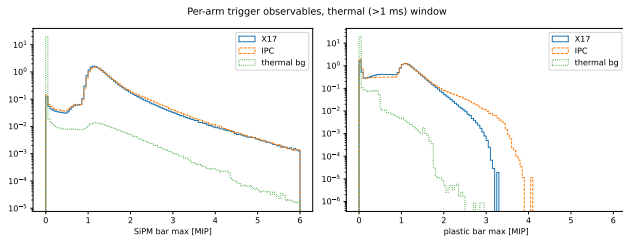
Run	primaries	purpose
thermal n	10^9 1 meV–2 eV	in-gate bkg
epi n	5×10^8 2 eV–100 keV	spill-in check
pairs	10^7 50/50	X17+IPC signal

Two physics details that matter

- **Self-shielding:** thermal captures sit within $\sim$ mm of the gas entrance (median 14 mm depth, 95% < 25 mm). Pair vertices are sampled from the *measured* capture profile, not uniform-in-gas ($\sim$ 3.7 cm upstream of centre).
- **Gate $\leftrightarrow$ energy:**
 $\text{TOF}[\text{ms}] = 1.41/\sqrt{E_n[\text{eV}]}$ at 19.5 m;
 $> 1 \text{ ms} \Leftrightarrow E_n < 2 \text{ eV}$, $4.28 \times 10^6 \text{ n/pulse}$.

Geant4 has no internal pair conversion — X17 and IPC are explicit generator modes; the neutron runs supply the capture- γ background and all singles.

Trigger observables — per-arm pulse height in MIP units



Signal legs peak at 1 MIP (crossing $e^\pm$); stopped legs reach ~ 3 MIP in the plastic.

Background (thermal captures):

- ▶ plastic hits die above ~ 1.7 MIP — the ^{28}Al 7.72 MeV Compton edge; H-capture 2.22 MeV γ stays < 0.5 MIP.
- ▶ SiPM shows no MIP peak (γ Comptons only).

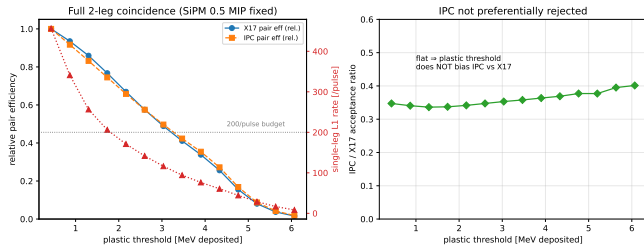
$\Rightarrow$ the **plastic** pulse height is the background-rejection lever.

Topology: full coincidence costs efficiency, but is clean

Pair trigger (both arms)	$\varepsilon(\text{X17})$	$\varepsilon(\text{IPC})$	bkg pair-tags/pulse
SiPM-only (2 arms ≥ 0.5)	15.0%	—	42
2 SiPM + ≥ 1 plastic confirm	7.0%	5.3%	1.8
full: $2 \times (\text{SiPM} \times \text{plastic})$	2.1%	0.7%	0.17

- ▶ Requiring the plastic on *both* legs costs $\sim 5\times$ in signal — pure solid angle (plastics are smaller and further back).
- ▶ **This study assumes the full coincidence** (the clean, high-purity choice), and optimizes the plastic threshold b within it.

Key result: the plastic threshold does *not* reject IPC



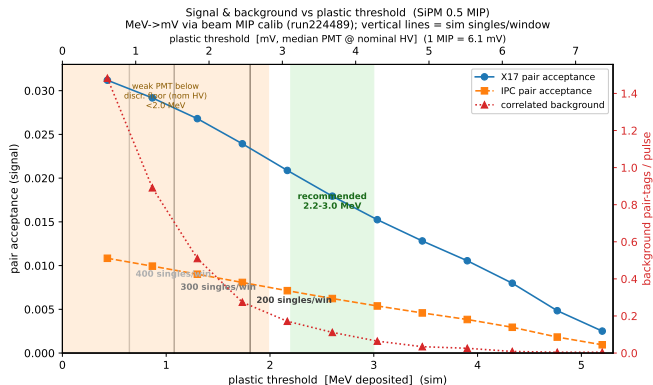
IPC/X17 acceptance ratio is **flat** (~ 0.35) across the whole threshold range.

Why: IPC's $1/M_{ee}$ spectrum favours soft, collinear pairs — 97% of triggering IPC has both legs in the *same* arm and is lost to the 2-arm requirement *on geometry*, before any energy cut.

The surviving IPC looks kinematically like X17, so b cuts them in lockstep.

You may raise the plastic threshold freely for background/rate control: it costs total statistics, **not** the IPC-vs-X17 balance.

Signal-to-noise optimization



Signal (left, blue/orange) falls *gently*; the correlated capture background (right, red) **collapses by ~ 1.5 MeV** once you clear the capture- γ Compton edges.

So S/N keeps improving with threshold — the cost is total statistics, not the IPC/X17 balance.

Vertical lines: the plastic threshold at which the (simulated) singles rate — the **OR of all four walls** — equals 200 / 300 / 400 per window, i.e. the DAQ capacity ceilings.

Sweet spot **plastic $\approx 2\text{--}3$ MeV**: background already < 0.1 /pulse, $\sim 50\text{--}65\%$ of X17 & IPC kept, IPC/X17 undistorted, singles under the 200/window ceiling, and it clears the hardware discriminator floor (orange, nominal HV) — see next slide.

Hardware anchoring — beam MIP + Y88 calibration

Sim response is validated

- ▶ Sim plastic MIP (Landau MPV)
= 4.33 MeV vs beam mean dE = 5.05 MeV
 $\Rightarrow$ ratio 0.86 = expected Landau MPV/mean for 2.5 cm PVT.
 - ▶ Sim SiPM-bar MIP 458 keV consistent with 0.3 cm PVT.
- $\Rightarrow$ thresholds quoted in **MIP transfer directly** to hardware.

DAQ landmarks (run224460/89, Y88 699/1612 keV edges): plastic MIP peaks span **3.3–10.3 mV** ($3.2\times$ gain spread); SiPM wall runs 12–14 mV ≈ 0.25 of its MIP-sum peak (consistent with the sim's insensitivity to the SiPM threshold).

Set **per-PMT** mV (not one global value) — the $3.2\times$ gain spread would otherwise give a $3.2\times$ spread in effective MIP threshold across arms. Equalize first via `pss/global_gain_slide` if a **uniform mV** is preferred.

Recommended 0.6 MIP $\rightarrow$ per-PMT threshold

PMT	mV	PMT	mV
PSSA1	6.2	PSSC1	3.9
PSSA2	4.8	PSSC2	3.8
PSSB1	2.2	PSSD1	3.0
PSSB2	3.6	PSSD2	2.0

Discriminator floor ≈ 1.5 mV $\Rightarrow$ the weakest PMT (PSSD2) cannot go below **~ 2.0 MeV (0.46 MIP)** at nominal HV — which is exactly where the S/N optimum sits, so no conflict. Lower thresholds need a plastic-HV raise.

Rate budget & the S/N table

plastic [MeV]	$\varepsilon(\text{X17})$ pair	$\varepsilon(\text{IPC})$ pair	corr. bkg /pulse	single-leg rate/pulse	
0.9	0.029	0.010	0.89	341	
1.7	0.024	0.008	0.27	206	← 200 budget
2.6	0.018	0.006	0.11	142	recommended
3.0	0.015	0.005	0.06	116	
3.5	0.013	0.005	0.03	94	MC floor

- ▶ **Measured DAQ capacity** (zs_ipd_safety, 2026-07-18): ~ 101 ev/window at IPD=10 (clean), ~ 189 at IPD=5, ~ 226 at IPD=2 — so the **200/window budget is real** (aggressive today; 300–400 awaits the network upgrade).
- ▶ Single-leg (L1) singles are the DAQ constraint — the *pair* coincidence is < 2 /pulse. If the 2-arm coincidence is formed in hardware, *b* can go as low as you like.
- ▶ **Singles caveat:** the vertical lines use *simulated* singles (~ 450 – 550 /window at the floor), $\sim 2\times$ below the DAQ-measured trigger level — so the true 200/300/400 thresholds sit somewhat higher. Re-anchor once a clean scaler-based singles-vs-threshold scan exists (M5 scaler was dead in the July run).

Summary & recommendation

- ▶ **Trigger:** full 2-arm SiPM×plastic coincidence, SiPM ≥ 0.5 MIP, **plastic $\approx 2.6\text{--}3.0$ MeV (0.6–0.7 MIP)**.
- ▶ **Keeps both signals:** the plastic threshold does not bias IPC vs X17 (flat ratio); it trades total statistics for purity.
- ▶ **S/N-optimal in the measurable range**, correlated background $< 0.1/\text{pulse}$, comfortably under the single-leg rate budget.

Caveats / next steps

- ▶ Thermal-gate signal production is tiny (self-shielding ceiling $\sim 10^{-4}$ IPC/pulse) — this window is background characterization; the physics ROI is the μs -scale MeV window (rates $\times 10^4$): re-run the same scan there.
- ▶ Accidental (uncorrelated) coincidences scale as (single-leg rate)²; full optimization needs the coincidence window — Python time-domain layer.
- ▶ If the *IPC invariant-mass spectrum* is a goal, keep b lower and correct with efficiency (a high b sculpts the accepted M_{ee} shape).